

DESIGN AND MAKE

The Make Phase

Group 48

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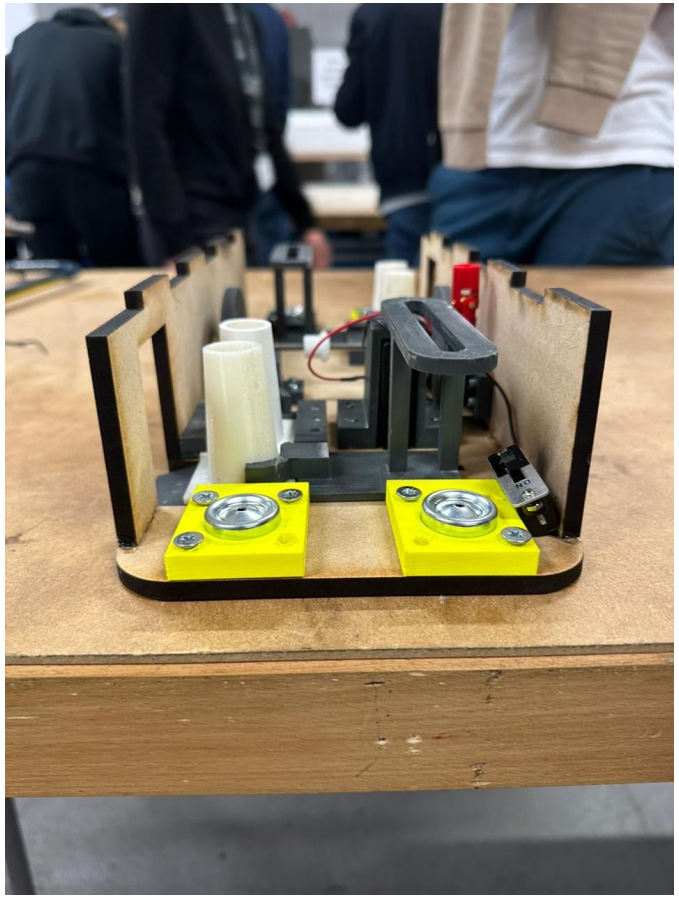
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INTRODUCTION

- The first phase of this project was completed in year 1, involving the design process of a vehicle that drops cones at periodic intervals.
- This year, the make phase was carried out to produce and assemble the designed prototype, testing its functionality against expectations.
- Initially, a consultation session took place to evaluate and validate the manufactured and ordered parts from year 1.
- The team assembled the prototype and tested its success on a later day



PROTOTYPE ASSEMBLY

PROTOTYPE ASSEMBLY PROCESS

Initial checks and revisions

- The team worked together on inspecting and assembling all parts, with each person taking responsibility on individual sections of the prototype.
- Initially, components inside the chassis were made sure to fit the cutouts on the chassis and ensured to work as intended.



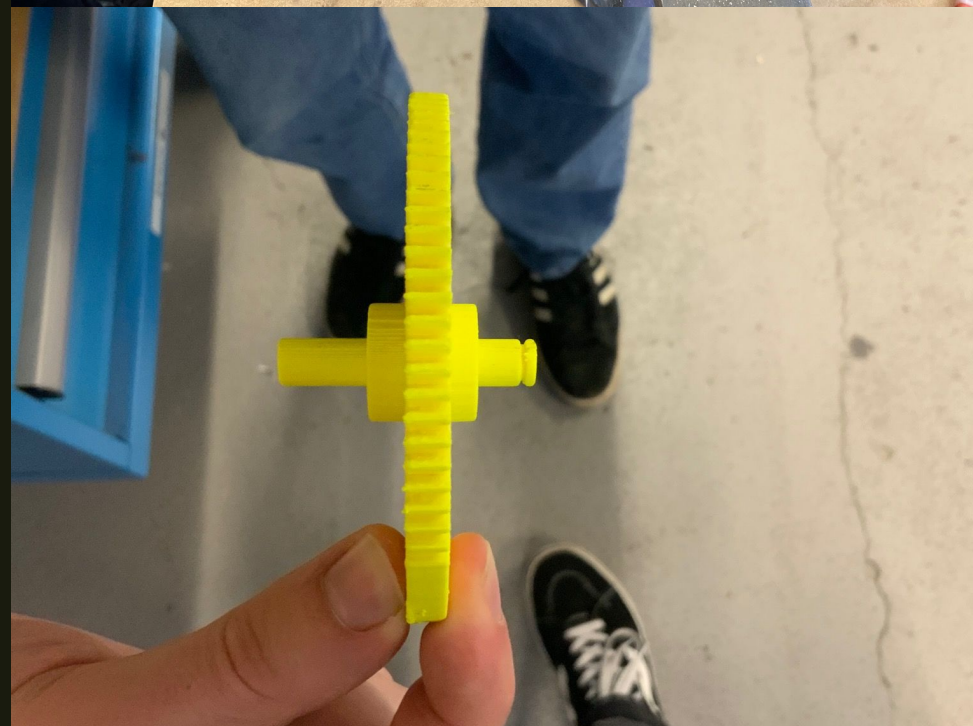


PROTOTYPE ASSEMBLY PROCESS

Initial checks and revisions

- The shaft of the yellow 3-D printed mutilated gears had to be sanded down so it would rotate with less friction, and so the pulleys would fit snug onto the gear.
- To aid the manufacture of the vehicle, holes were drilled all the way through components to ensure screws would not fall out.
- The 3D-printed parts were not initially tapped; however, by screwing in the screws, the softness of the plastic resulted in the creation of threaded holes.
- Following this, shafts were cut to the required sizes using a saw.





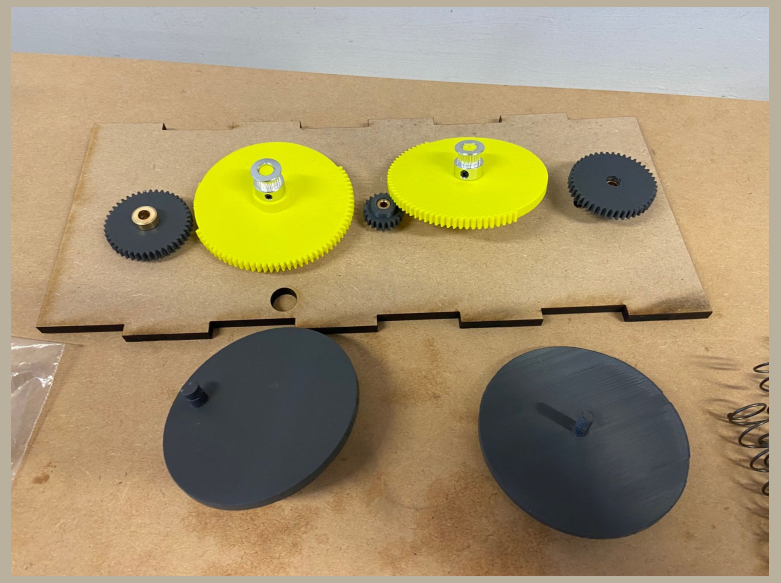
PROTOTYPE ASSEMBLY PROCESS

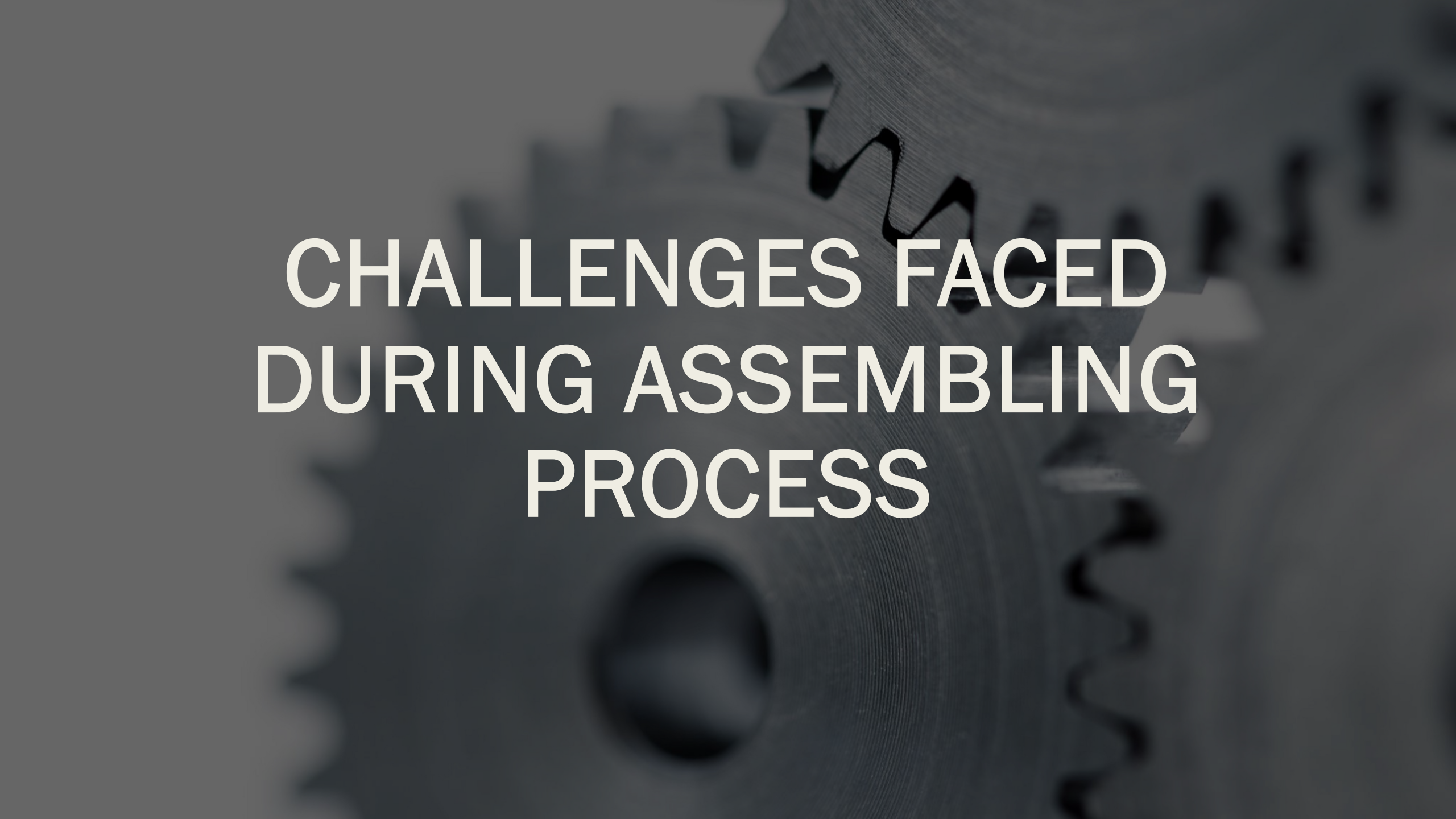
Assembly

- Once all components were verified, the 3D printed parts were fit onto the chassis, including the sides, with screws.
- Following this pulleys were installed onto each of the gears, and the gear mechanism was put together, constrained with circlips to the vehicle roof.
- Each of the four roller bearings were installed in each corner of the chassis to support the prototype in movement.
- Next, the wheels were inserted into the shaft to provide further support.
- An iteration to the design, screw countersinks were installed to ensure screws were flush with the chassis, specifically to avoid them touching the ground and inhibiting the vehicles movement.



(photo showing ball transfer units)





CHALLENGES FACED DURING ASSEMBLING PROCESS

Challenges

Sliders and Gears

- *The Problem*: Friction appeared when making the sliders glide

The Solution: Sliders underwent sanding to minimise friction, enhancing their ability to smoothly glide.

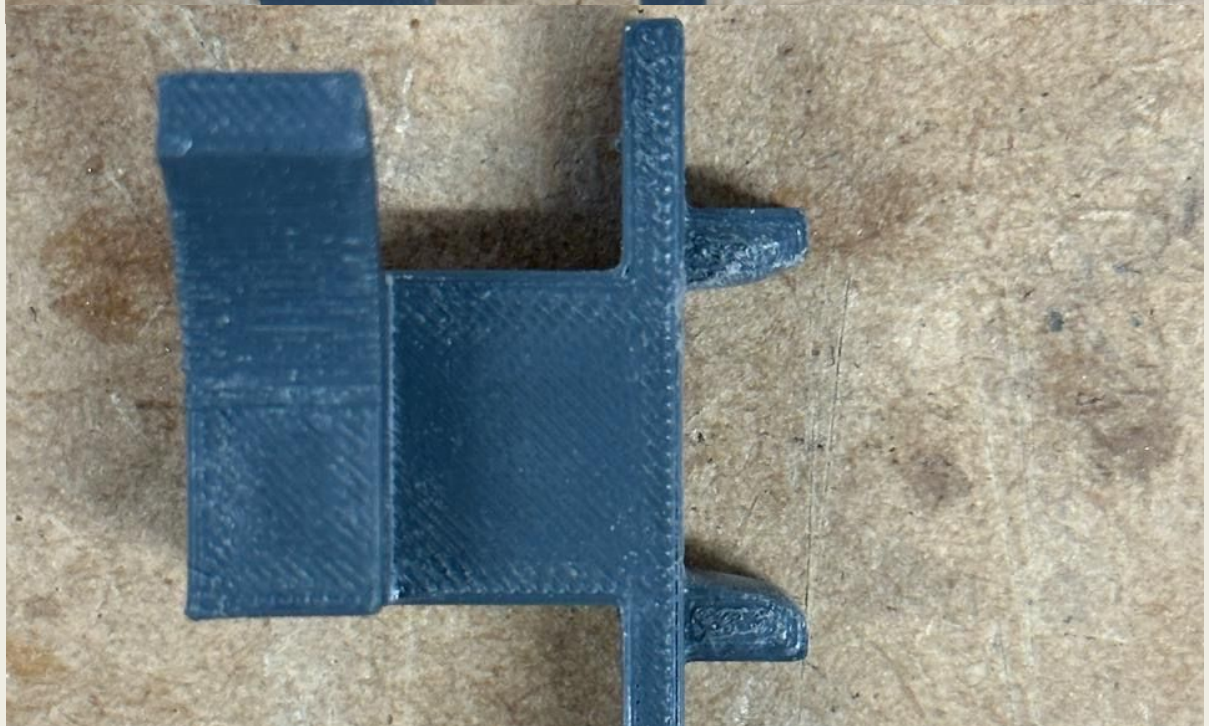
- *The Problem*: Friction appeared when couplings were contacting pulleys.

The Solution: The gears' outer surfaces were sanded to facilitate the insertion of couplings and pulleys.

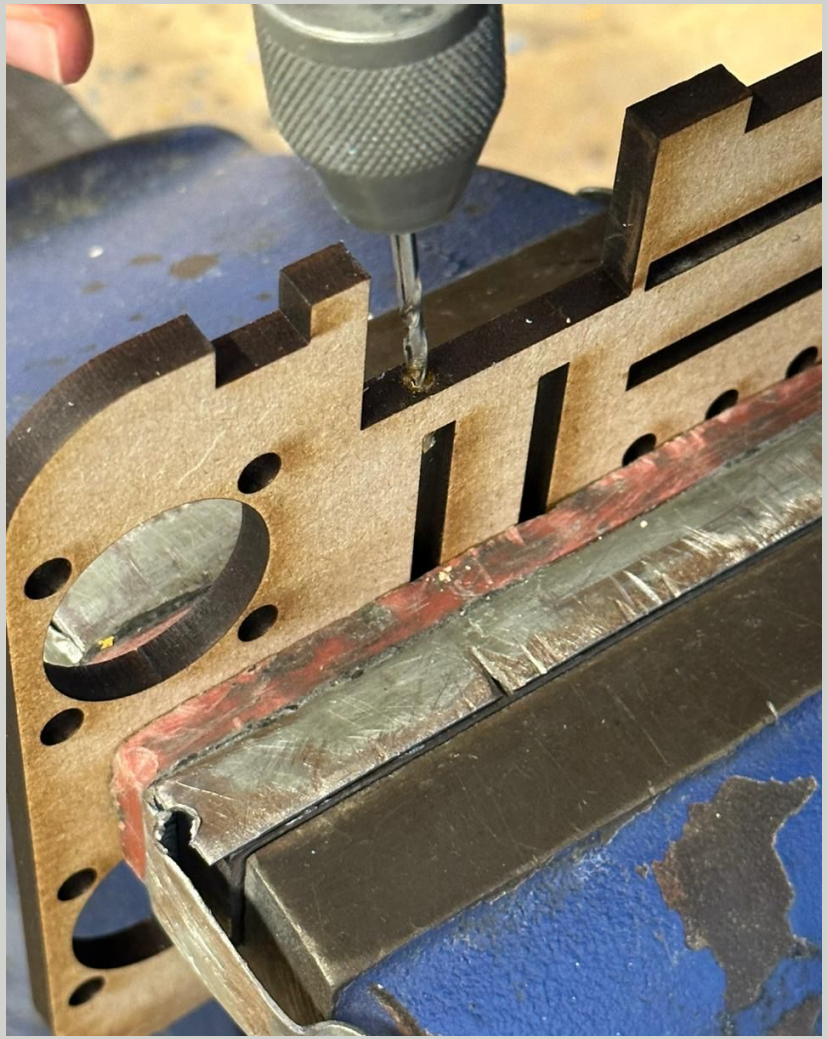
- *The Problem*: The scotch yoke arm was prevented the ejection to happen.

The Solution: One half of each scotch yoke arm was sawed to create more space within the busy vehicle for the cones to move into the desired position for ejection.

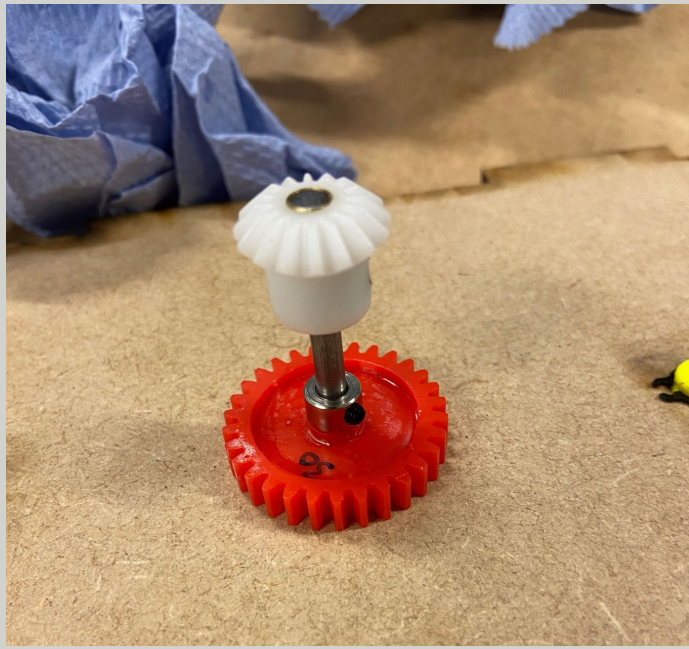




Challenges







THE PROBLEM: A break in the chassis bottom occurred when drilling holes for dowel pins, that are connecting the ramp to the chassis bottom.

THE SOLUTION: The break in the chassis bottom was filled with glue to secure the ramp in place.

THE PROBLEM: The bevel gears faced engagement issues due to shaft movement to the side.

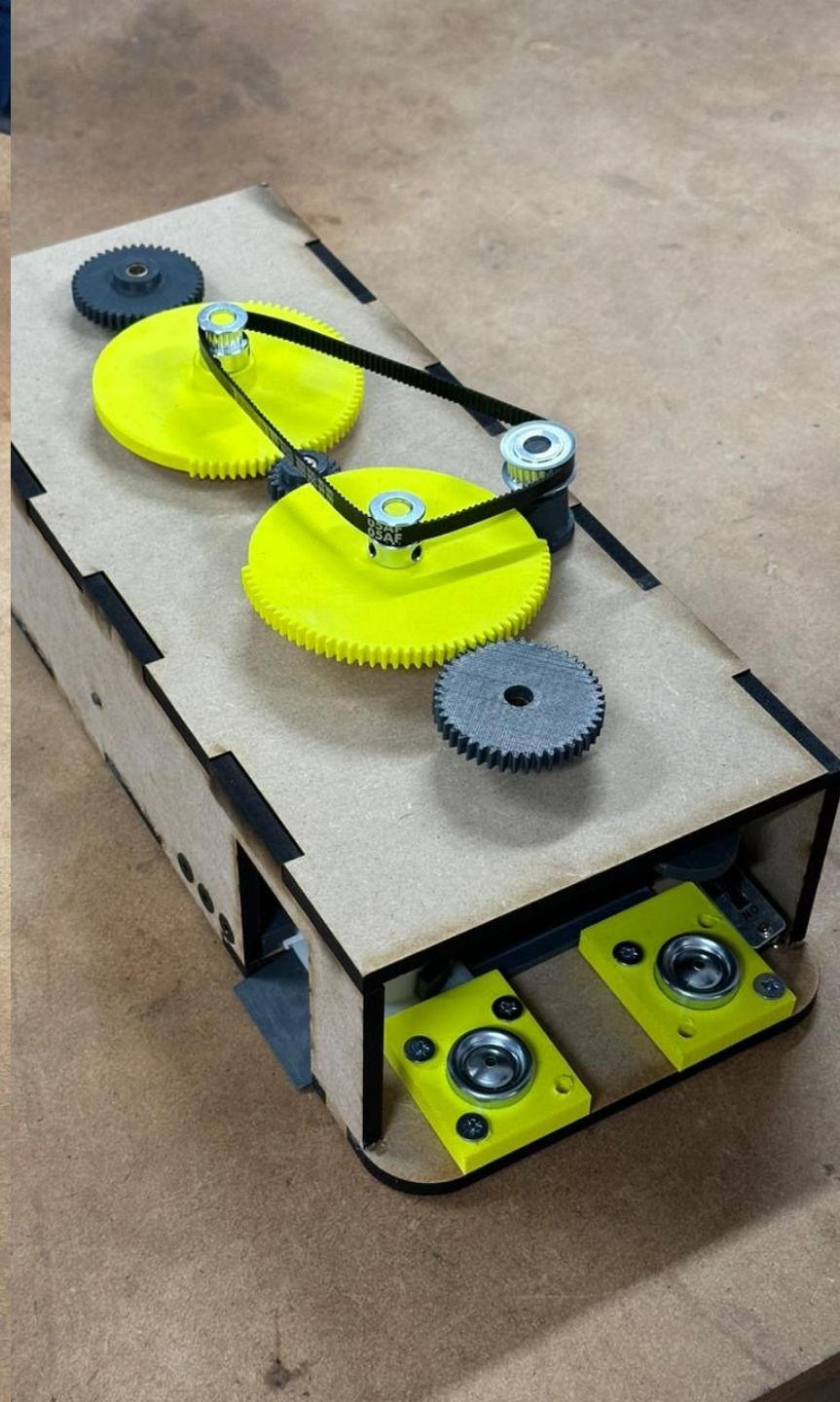
THE SOLUTION: To prevent this, a gear with a wide base was introduced to eliminate lateral motion.

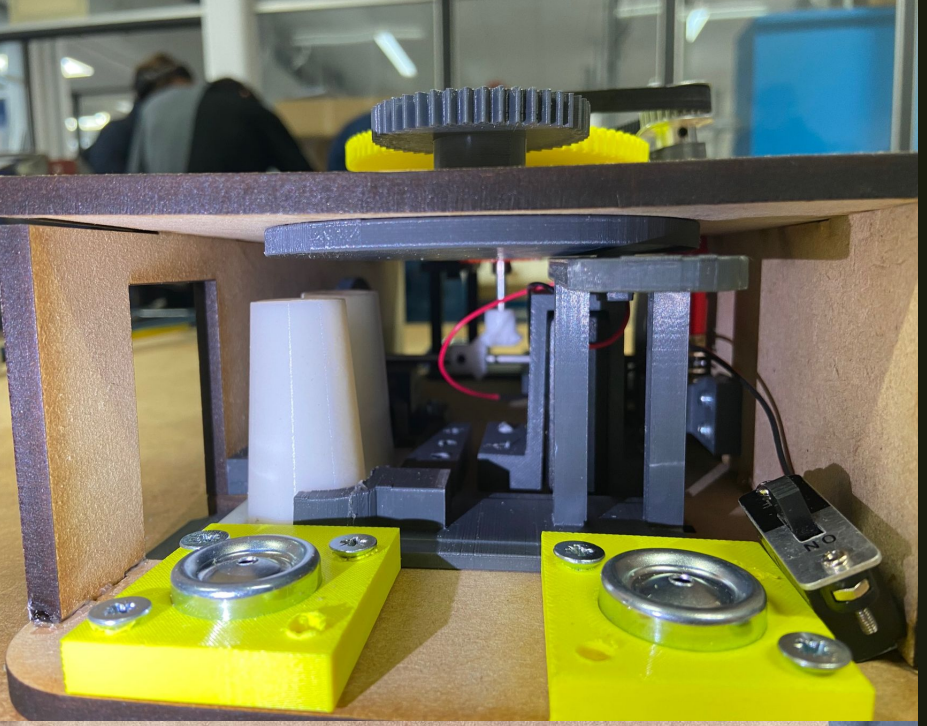
Challenges

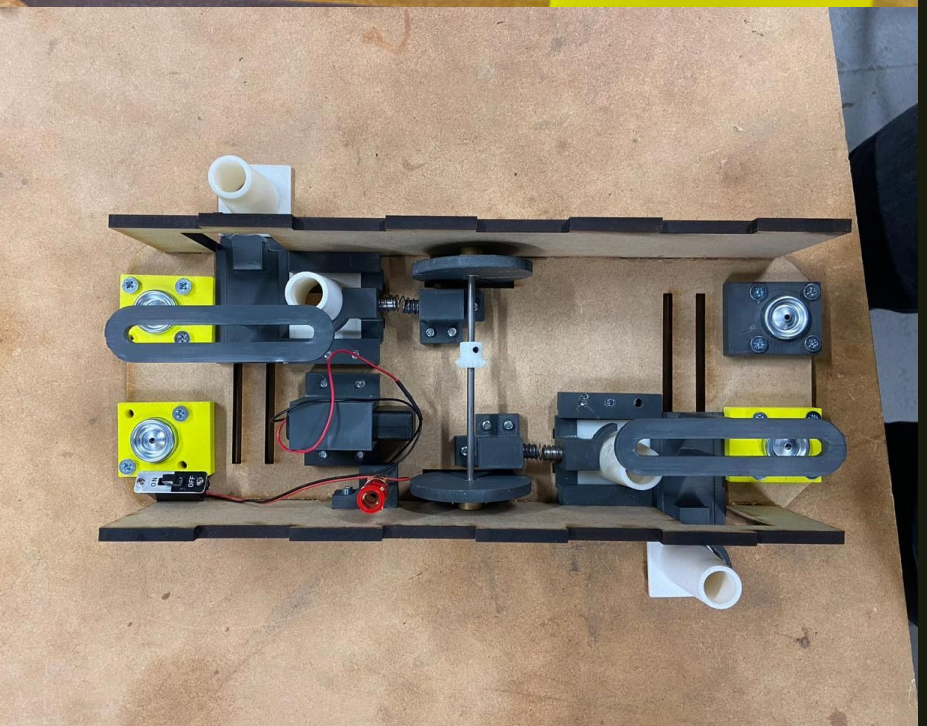
THE PROBLEM: The belt is slipping, so there's an issue with belt sizing.
THE SOLUTION: Despite changing the motor pulley to a larger size, the problem persisted.

THE PROBLEM: Wheels are slipping when in contact with the floor.
THE SOLUTION: Rubber bands were applied to the wheels, elevating the vehicle's ride height to ensure proper ground clearance. This adjustment also led to an improvement in traction.

THE PROBLEM: The chassis components would not hold together as the finger joints were too wide
THE SOLUTION: Glue was used to secure them in place.







Challenges

THE PROBLEM: The failure of screw countersinks to secure properly resulted in insufficient wheel contact with the ground.

THE SOLUTION: To remedy this issue, rubber band tires were employed to enhance height and traction

THE PROBLEM: Concerns arose about turning on the machine for the test run due to the cramped design.

THE SOLUTION: Consequently, a switch was wired into the system to address this issue.

THE PROBLEM: The spring employed for the cone magazine proved excessively powerful and rigid, leading to rapid and, at times, problematic pushing of the replacement cone, resulting in major jams.

THE SOLUTION: A less rigid spring was deemed necessary, but none were available.

IMPROVEMENTS AND LEARNING OUTCOMES

The build session went well overall, and we were able to overcome most issues that were encountered. However, there are many improvements that could be made to improve the design and make the build session more efficient.

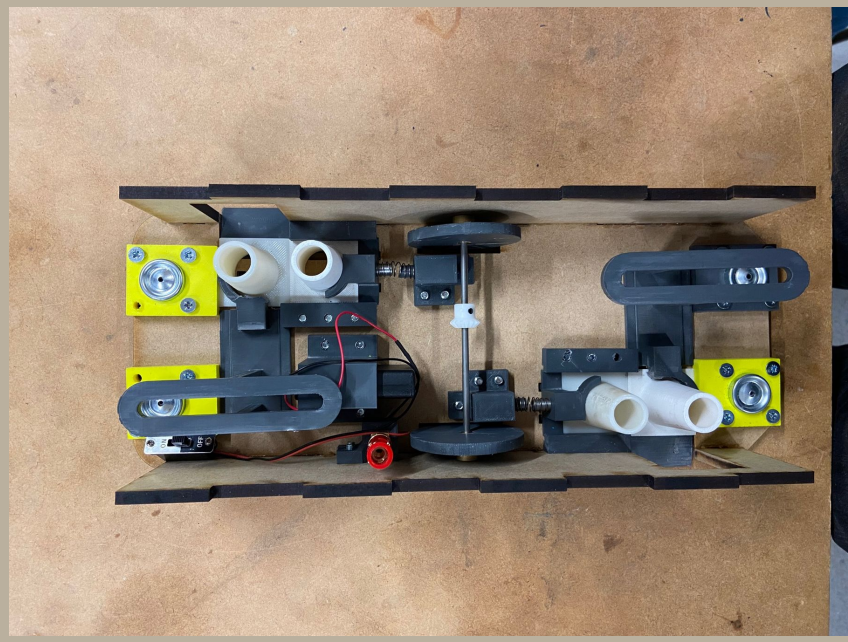
DESIGN IMPROVEMENTS

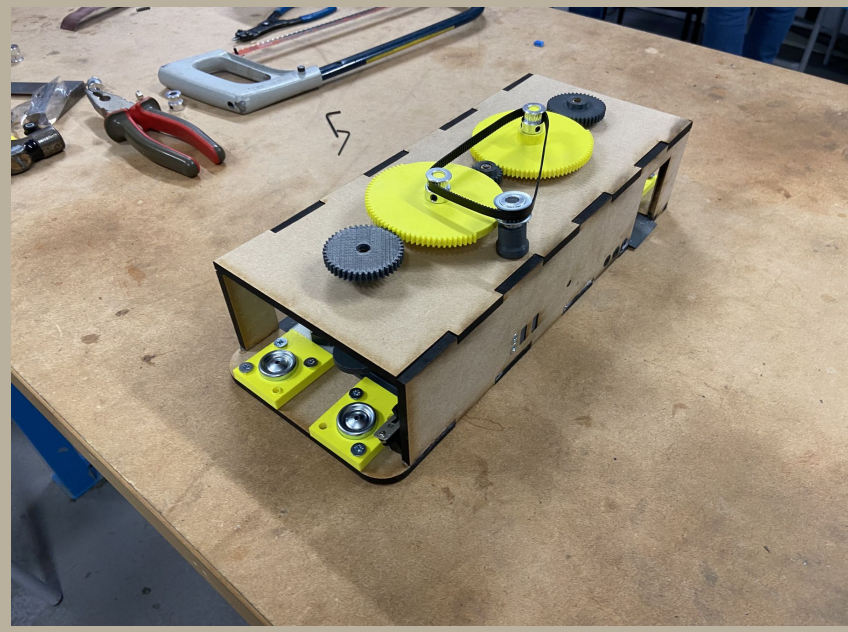
Specific Issues

- Scotch Yoke Mechanism issue:
  - ❑ Problems: Unreliable performance due to height and friction problems
  - ❑ Solution: Sand 3D components, apply lubricant on interacting surfaces to improve functionality

- Thin cone rails issue:
  - ❑ Problem: Inadequate prevention of cone toppling when pushed
  - ❑ Solution: Lengthen rail overhang to enhance stability

- Spring in cone magazine issue:
  - ❑ Problem: Frequent jams
  - ❑ Solution: Consider a simpler feeding ramp mechanism





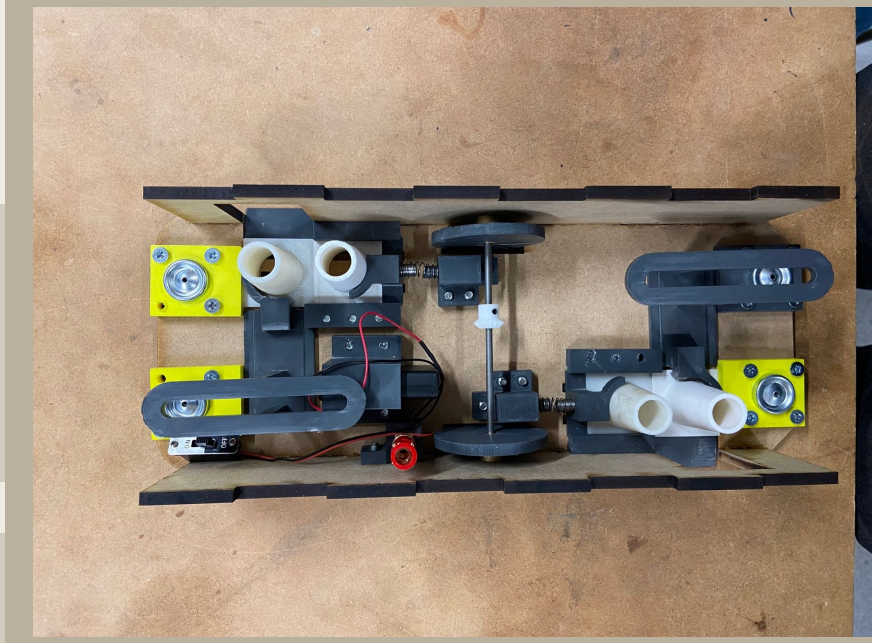
DESIGN IMPROVEMENTS

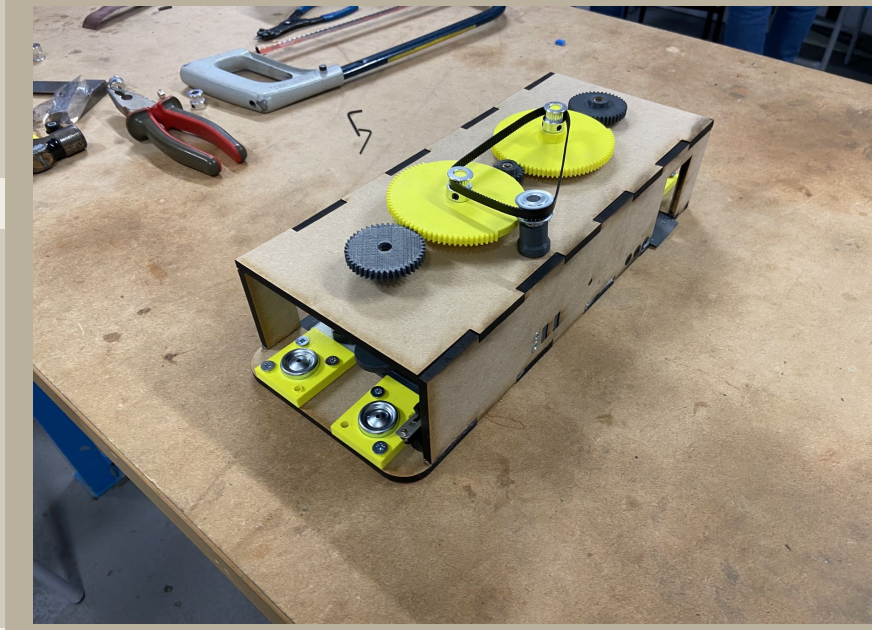
Specific Issues

- Oversized holes in MDF chassis:
  - ❑ Problem: Screws grip issues due to large holes
  - ❑ Solution: Use larger screws or create a new part with appropriately sized holes

- Chassis assembly challenges:
  - ❑ Problem: Insufficient friction in finger joints, necessitating glue for stability
  - ❑ Solution: Consider using dovetail joints for improved joint friction and assembly

- Belt slippage on pulleys
  - ❑ Problem: Pulley bending under belt tension leading to slippage
  - ❑ Solution: Extend shaft, add a third horizontal chassis piece for secure gear mounting to counteract belt tension

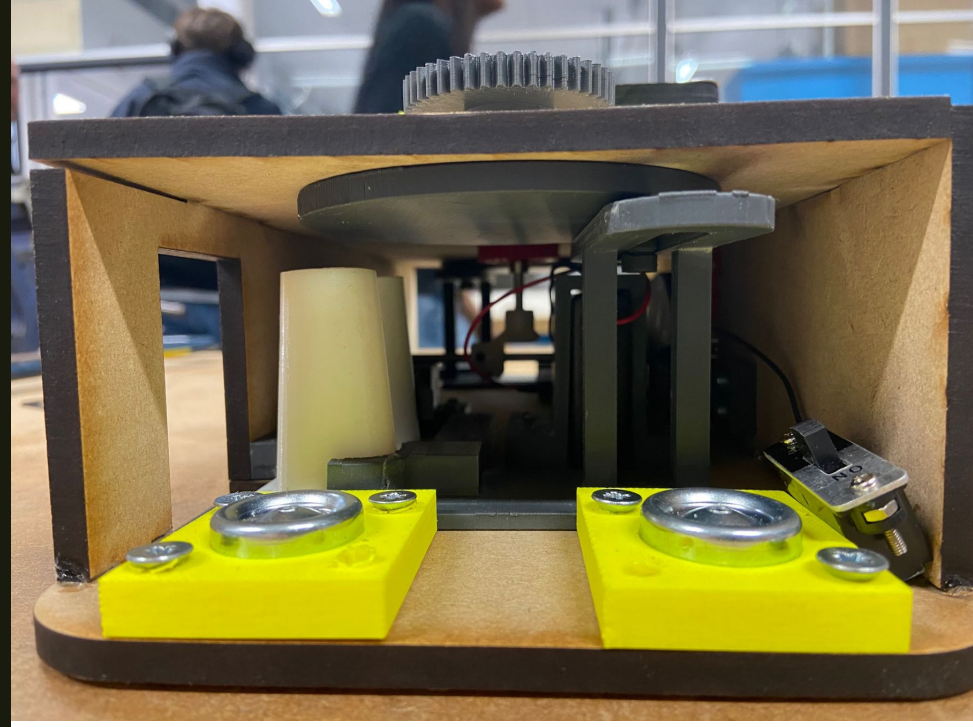


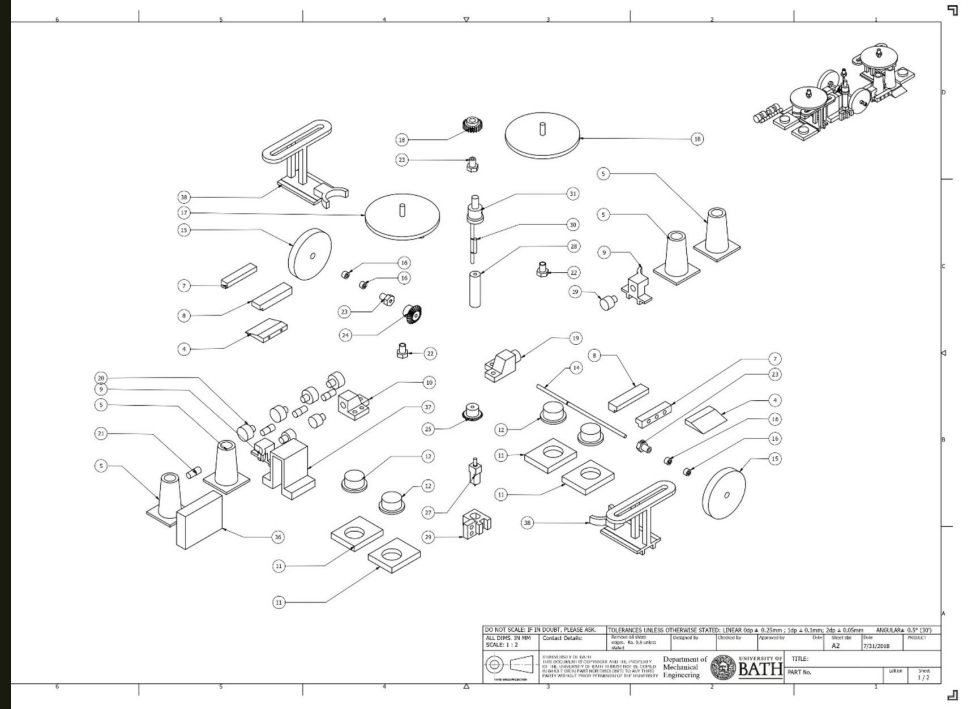


DESIGN IMPROVEMENTS

Overall issues

- The design could have been simplified to reduce the number of components and potential failure points.
- A different cone pushing mechanism could have been chosen as a scotch yoke mechanism is complicated to execute without precise manufacturing and multiple prototyping stages.
- Less 3D printed parts could have been used, mainly for the scotch yoke, wheels and gears, in favour of smoother and more precisely tolerance acrylic components or off the shelf components.

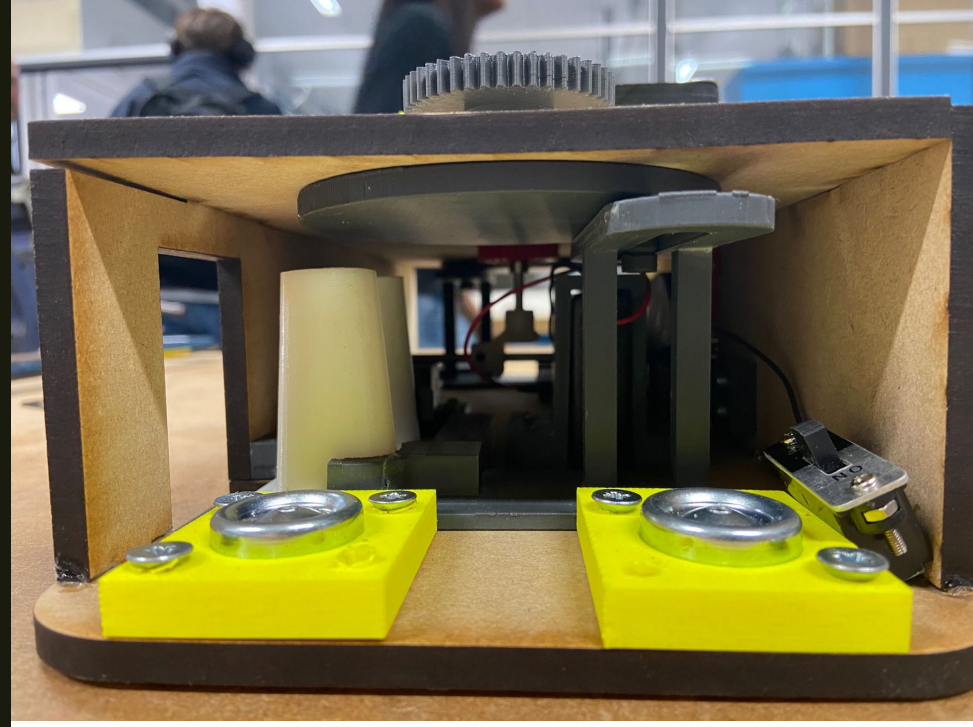


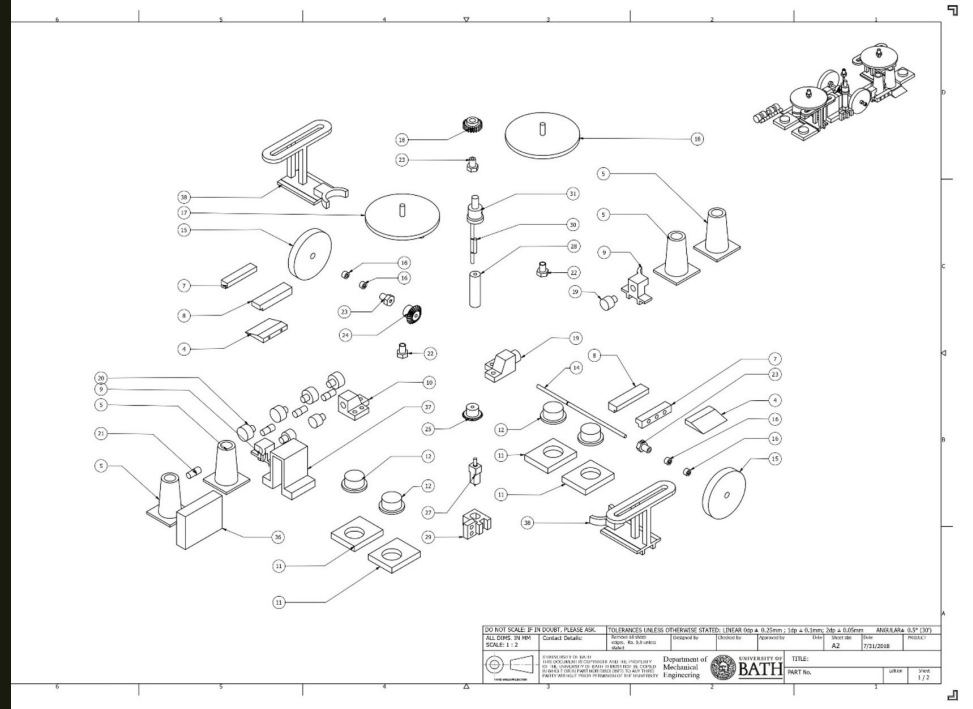


DESIGN IMPROVEMENTS Overall issues

- The main body could have been made from acrylic to improve strength, tolerancing and visibility inside the machine.

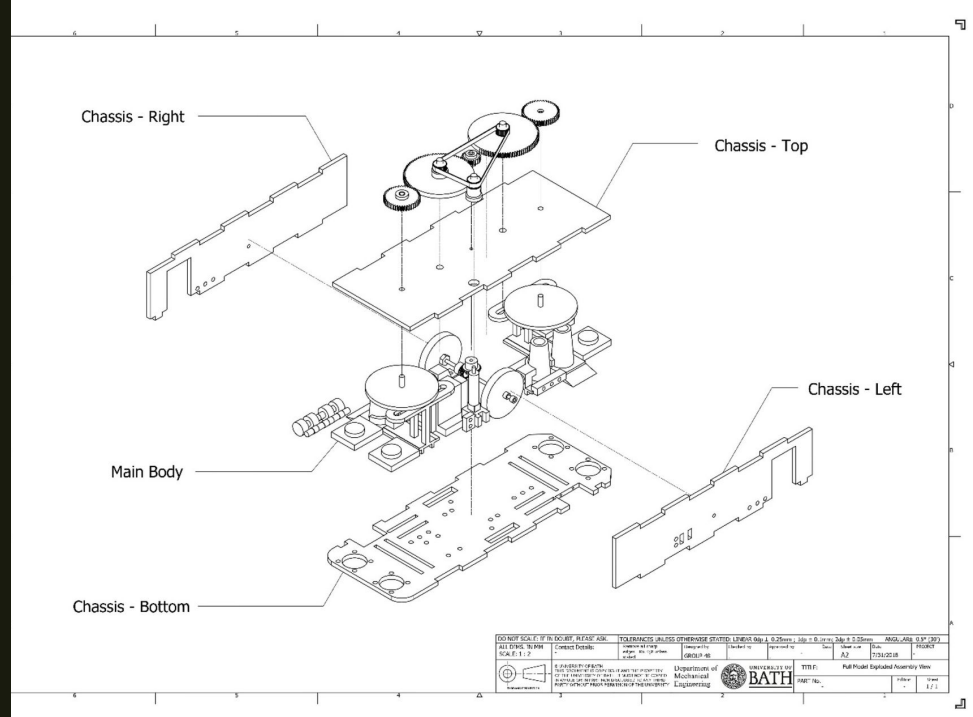
- Due to the design, to re-load cones into the vehicle, the roof had to be removed, which involved taking apart the gear mechanism. Re-attaching all components correctly was tricky, especially meshing the bevel gears together.



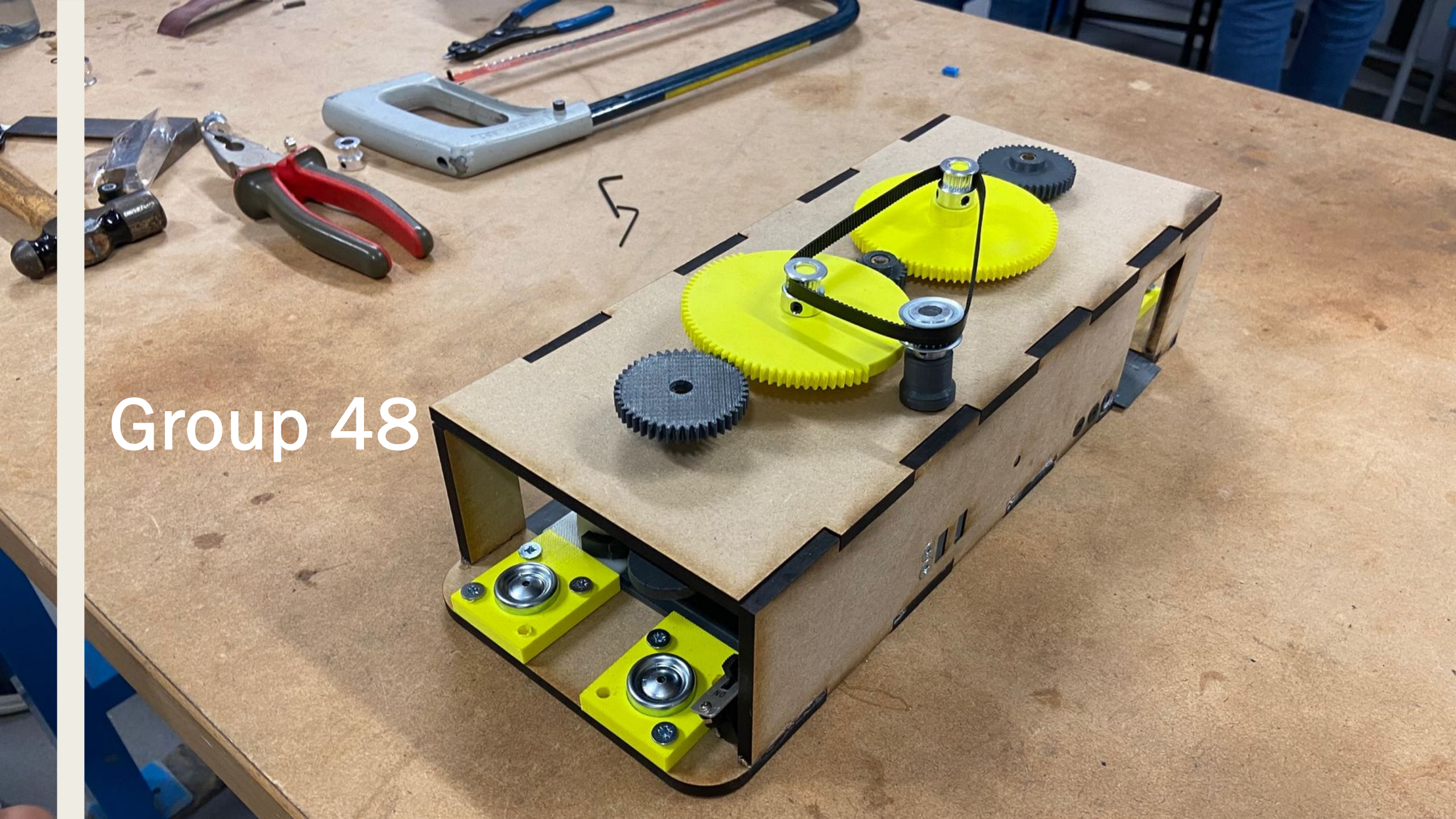


BUILDING PROCESS IMPROVEMENTS

- More planning before the build session could have made the session run more smoothly improving role allocation, component preparation and assembly process
- A small meeting to review the mechanics of the design and its assembly could have been scheduled to re-familiarise the team with the design before building it






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